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Article in Journal of Asian Natural Products Research · August 2024

DOI: 10.1080/10286020.2024.2390510

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To cite this article: Abdullah Al Mamun, Khorshed Alam, Farjana Akter Koly, Farjana Showline Chaity, Jannatul Ferdous & Saiful Islam (14 Aug 2024): Genome mining for ribosomally synthesized and post-translationally modified peptides (RiPPs) in *Streptomyces* bacteria, Journal of Asian Natural Products Research, DOI: [10.1080/10286020.2024.2390510](https://doi.org/10.1080/10286020.2024.2390510)

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Genome mining for ribosomally synthesized and post-translationally modified peptides (RiPPs) in *Streptomyces* bacteria

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ABSTRACT

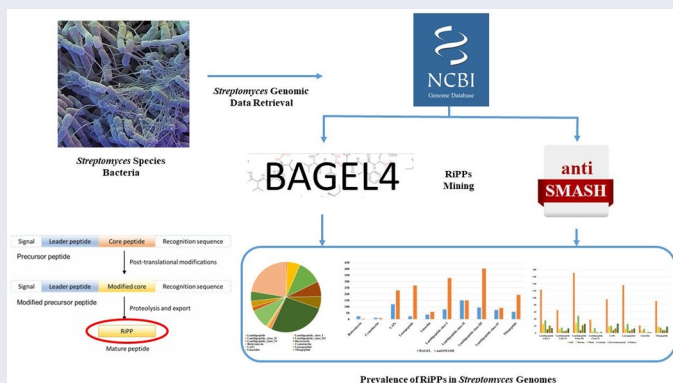
Ribosomally synthesized post-translationally modified peptides (RiPPs) are a novel category of bioactive natural products (NPs). *Streptomyces* bacteria are a potential source of many bioactive NPs. Limited opportunities are available to characterize all the bioactive NP gene clusters. In this study, 410 sequences of *Streptomyces* were analyzed for RiPPs through genome mining using the National Center for Biotechnology Information (NCBI), by combining BAGEL and anti-SMASH. A total of 4098 RiPPs were found; including both classified (lanthipeptide, RiPP-like, bacteriocin, LAPs, lassopeptide, thiopeptides) and nonclassified RiPPs. Soil was identified as a rich habitat for RiPPs. These data may offer alternative future remedies for various health issues.

ARTICLE HISTORY

Received 11 December 2023
Accepted 2 August 2024

KEYWORDS

Genome mining;
Streptomyces; RiPPs;
natural products;
antibiotic resistance



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 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/10286020.2024.2390510>.

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1. Introduction

Antibiotics are like a gift to humanity because they can stop bacterial infections, but when they are overused and lead to antibiotic resistance, they become a burden. It is imperative to develop new antimicrobial medications to combat the expanding global issue of antibiotic resistance. Antibiotic therapy is also the most crucial strategy for battling or preventing bacterial illnesses in animal populations and in the food business [1]. Despite the success of the discovery of new antibiotics and improvements in the manufacturing process, infectious diseases continue to be the second leading cause of death worldwide. Bacterial infections account for about 17 million of those deaths each year, mostly affecting the elderly and young children [2].

Over the past century, natural products (NPs) have significantly advanced our understanding of biology and contributed to the advancement of medicine. A recent study found that NPs or their compounds made up more than half of all authorized drugs from 1981 to 2019 [3]. Terpenoids, alkaloids, polyketides, and non-ribosomal peptides were four of the NPs classes that were discovered through research in the twentieth century [4]. The world needs a brand-new class of antimicrobials to combat the rising rate of antibiotic resistance before it becomes a global epidemic.

With the discovery of streptothricin in 1942, the history of antibiotics derived from *Streptomyces* began [2]. Two years later, with the discovery of streptomycin, researchers stepped up their quest for antibiotics within the species. Actinomycetes, the most significant subgroup of the *Streptomyces* genus, are the source of 80% of the antibiotics used today [2].

Streptomyces, a genus of Gram-positive bacteria, develops in a variety of conditions and has a filamentous shape like fungi [2]. More than 900 species in the genus *Streptomyces*, which belongs to the family streptomycetaceae in the actinomycetota phylum, offer appropriate chassis for NP discovery. They can be found in a variety of settings, but the soil is where they are most prevalent, accounting for 1 to 20 percent of all living things, and it is thought that they like the distinctive earthy odor (geosmin) [5]. This genus' most prominent quality is its capacity to produce secondary metabolites. Many of these secondary metabolites have valuable industrial and pharmaceutical applications, and there is also a lot of structural variety among these secondary metabolites, which include antibacterial, antifungal, antiviral, anti-hypertensive, immune suppressant, and antitumor compounds [6]. More than 6000 distinct compounds and approximately 75% of antibiotics used both therapeutically and economically are produced by *Streptomyces* species (e.g. neomycin, cypemycin, grisemycin, bottromycins, and chloramphenicol) [7].

Genome mining-based approaches have emerged as potent and increasingly well-liked techniques for the automatic discovery of biosynthetic genes to identify and annotate putative biosynthetic gene clusters (BGCs), frequently utilizing algorithms based on the understanding of the NP biosynthetic machinery and the biosynthetic potential of microorganisms [8]. Sequencing of *Streptomyces* genomes has revealed that the majority of *Streptomyces* genomes contain 20 or more BGCs for secondary metabolites, with the vast majority of these metabolites being previously unsuspected and uncharacterized and absent from most other strains [9–11].

Ribosomally synthesized and post-translationally modified peptides (RiPPs) have also been demonstrated to be somewhat more efficient antibiotic replacements than the other three NP families because of their structural diversity and functional variety [12]. The first decade of the twenty-first century's genome sequencing efforts has shown that RiPPs comprise another important class [4]. RiPPs are a subclass of peptides with post-translational modifications (PTMs) that enhance the complexity of their chemical and structural composition [13]. RiPPs biosynthesis starts with the expression of a precursor peptide through the typical ribosomal route. Research indicates that the genetic basis for RiPPs production is now well recognized and that there are often specific conserves [14]. RiPPs have a restricted spectrum, unlike traditional antimicrobials, and show multiple mechanisms of action simultaneously, which signifies they can stop DNA, RNA, or protein biosynthesis and have minimal chance of developing resistance because their specific targets are mainly conserved and do not support significant changes [15].

Actinomycetes, like the *Streptomyces* species, which have between 20 and 40 BGCs, are examples of bacteria with a highly specialized, complex metabolism [13]. The NCBI datasets contain a large number of *Streptomyces* genome sequences, which is a useful resource for studies on the metabolic activities of these bacteria. In this study, we investigated all publicly accessible whole reference genome sequences of species in the *Streptomyces* genus using a range of bioinformatics tools to determine the prevalence of RiPPs and diversity and distributions across these genomes.

2. Methodology

2.1. Genome extraction

We used the NCBI Dataset's genome browser (NCBI: <https://www.ncbi.nlm.nih.gov/datasets/genomes/>, accessed on 20 April 2024) to search for and collect the *Streptomyces* complete genome sequences. We found a totally of 2889 different types of *Streptomyces* genomes, including contigs, scaffolds, chromosomes, and complete genomes. We filtered, as reference genome, an annotated and complete assembly level to obtain *Streptomyces* genome sequences and retrieved 410 complete reference genome sequences in FASTA format of different *Streptomyces* species from NCBI datasets on 20 April 2024. [Supplementary File-SF 1](#) show genome assembly, accession numbers, and genome information (genome size, gene number, and genes of protein-coding).

2.2. Genome mining

Using the two methods, we determined the number of RiPPs for each genome. To possibly find RiPPs involved in the synthesis of secondary metabolites, the genome mining prediction platforms antiSMASH 6.0 [16] and BACteriocin GENome mining tool (BAGEL4) [17] were applied. These platforms used a combination of computer tools with default parameters. The antiSMASH tool makes it simple to locate, identify, and research gene clusters involved in secondary metabolite production all over the genome. In a similar vein, BAGEL4 is designed to thoroughly mine RiPPs and

bacteriocin [16]. These advanced computer modeling services enable the precise prediction of the encoding potential of microbial secondary metabolites [13]. The principles of the hidden Markov model (HMM) [18], the BLAST algorithm [19], PFAM [20], GenBank [21], UniProtKB [22], BACTIBASE CAMPR3 [23], and the MIBig data repository [24] are only a few of the database systems that these programs use for RiPPs annotation from genomic sequences.

One of the earliest tools created, BAGEL4, allows researchers to locate and visualize gene clusters in bacterial DNA that are involved in the biosynthesis of (unmodified) bacteriocins and ribosomally produced and post-translationally modified peptides (RiPPs). BAGEL was first released in 2006, and several updated versions are available for free online. Better than its precursor, BAGEL4 is now totally independent of ORF-calling [17], and the updated version BAGEL4 is available for free online (BAGEL4: <http://bagel4.molgenrug.nl>, accessed on 13th October 2021). It was developed to address the issue of complex ORF annotation due to low sequence homology. BAGEL4, whose annotation database was upgraded with improved RiPP protein domain information, was the program's most current upgrade. The BAGEL website provides information on almost 500 RiPPs, 230 unmodified bacteriocins (class 2), and 90 large (>10 kD) bacteriocins in addition to BGC identification (class 3). A link to NCBI or UniProt is present in the majority of the data [16]. The researchers can perform a search against this database using a user-defined collection of precursor peptide sequences using the BAGEL4 “Core Peptide Blast” tool [13].

More web servers, such as antiSMASH, have been built after the BAGEL [17] first version was launched. A genome mining technique called antiSMASH (antibiotics and secondary metabolite analysis shell) is used to identify and analyze 52 different types of NP BGC. It was initially published in 2011 [16] and has since received numerous updates [25–29]. The public version of antiSMASH 6.0 can be found online (antiSMASH: <https://antiSMASH.secondarymetabolites.org/#!/start>, accessed on 13th October 2021), while R&D versions can be found online (R&D versions: <https://bitbucket.org/antiSMASH/>, accessed on 13 October 2021) [30]. antiSMASH is the most widely used tool for genome mining, with about 1335383 operations completed online as of this writing. antiSMASH mining is optimized for both fungal (fungiSMASH) [27] and plant (plantiSMASH) [30] genomes in addition to bacterial genomes. antiSMASH compares encoded gene products with a manually curated library of HMMs that lists several NP biosynthesis genes to perform its analysis. BGCs are discovered by designating specific metabolite-specific clusters of orthologous groups to essential enzymes in a gene cluster. The annotation of auxiliary genes, the prediction of BGC borders, and the prediction of compound final structures are all carried out as further downstream investigations. Comparing detected BGCs with known BGCs in the MIBiG database is made possible by the integrated Known ClusterBlast function. Numerous capabilities that offer a thorough RiPP BGC annotation have been added to antiSMASH over time while not specifically focusing on RiPPs [16, 25].

3. Results

This work investigated the occurrence of RiPP BGCs using complete and published genome sequences of *Streptomyces*. Out of 410 strains, 29 showed no hits according

to BAGEL genome mining tools and 9 showed no hits according to antiSMASH genome mining methods. In contrast, 336 out of 410 strains display 1 to 5 hits in the BAGEL genome mining tool. However, only 78 out of 410 strains in the antiSMASH genome mining tool yield more than one hit of RiPPs. However, the BAGEL genome mining tool indicates that 44 out of 359 strains have 6 to 10 hits; in contrast, the antiSMASH genome mining tool shows that 287 out of 359 strains have hits. Using the BAGEL genome mining tool, no strain was identified to have hits more significant than 10 RiPPs; however, using the antiSMASH genome mining tool, we discovered 37 strains with hits greater than 10 RiPPs ([Supplementary File-SF2](#)).

According to the BAGEL genome mining tool, it has detected eight classified RiPPs for instance, lanthipeptide, bacteriocin, bottromycin, cyanobactin, LAPs, lassopeptide, linaridin, and thiopeptide. Other types of RiPPs have also been seen, denoted as NC (Not Classified). Among the classified RiPPs, the distribution of bacteriocin is the highest, with almost 354 hits. Lanthipeptides class II and LAPs also give more than 100 hits, while the rest of the RiPPs classes showed less than 100 hits, whereas only eight lanthipeptides have been detected, which is the lowest. In the case of nonclassified RiPPs, a significant amount of hits (298) have been generated, which is valuable information for further analysis of nonclassified RiPPs ([Figure 1](#), [Supplementary File-SF 3](#)).

According to the antiSMASH genome mining tool, all classified RiPPs have been detected with a noteworthy amount of hits. antiSMASH was able to identified 14 types of classified RiPPs in all the genome sequences such as bottromycin, cyanobactins, guanidinotides, lanthipeptides, LAPs, lassopeptides, linaridins, lipolanthine, ranthipeptide, redox-Cofactor, RiPP-like, RRE-containing, thioamitides, and thiopeptides. Lanthipeptides showed the most prevalence hits of distribution in all the strains of *Streptomyces*, which has been used in this study, whereas lanthipeptides

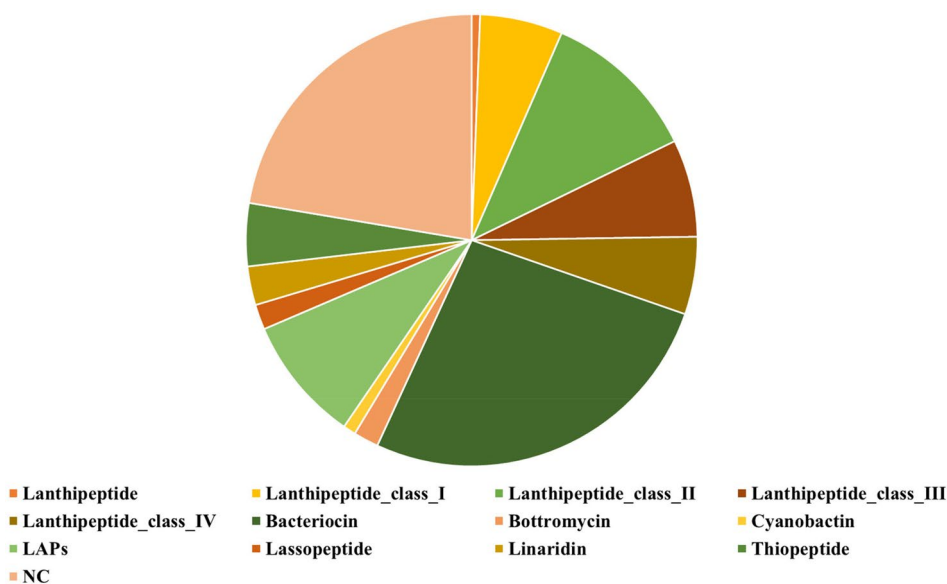


Figure 1. Distribution of different RiPPs (BAGEL hits).

class III appearance is higher, with 402 hits. Most of the classified RiPPs denoted to give more than 100 hits, such as RiPP-like, Lap, lasso peptide, Redox-Cofactor, RRE-containing, and thiopeptides. Even though the rest of the RiPPs types give fewer hits, they are also valuable findings (Figure 2, Supplementary File-SF 4).

In comparison, almost ten types of RiPPs have been exposed as similar classes of RiPPs hits between both of the genome mining tools such as Bottromycin, Cyanobactin, LAPs, Lasso peptide, Linaridin, Lanthipeptide class I, Lanthipeptide class II, Lanthipeptide class III, Lanthipeptide class-IV, thiopeptide. antiSMASH gives more hits than BAGLE except for bottromycins and cyanobactins. Differences between antiSMASH and BAGEL are very high, which can be particularly seen in Figure 3. Lasso peptide, Lanthipeptide class I, Lanthipeptide class III, and thiopeptide give almost three times higher hits in the case of the antiSMASH genome mining tool. But in the case of Lanthipeptide class II, BAGEL gives only one hit more than the antiSMASH tool, which seems to have an equal number of RiPPs hits (Figure 3).

4. Data based on sources

Based on the habitats, six different sources have been denoted: soil, marine, plant, animal, environmental (soil excluded), and others. It is expected that soils are most likely to produce RiPPs. After that, plants are the potential source of a significant amount of RiPPs, which give more than 200 hits. Though other sources contain several habitats, marine habitats provide almost similar kinds of hits of RiPPs. Due to excluding the soil sources from environmental habitats, RiPPs have been identified in meager amounts, which might be higher than in animal habitats. Still, compared to soil habitats, they are almost five times lower (Figure 4, Supplementary File-SF 5).

A complete and detailed scenario is observed in the case of individual RiPPs classes in the listed habitats used for this study. Selective classes of RiPPs have been presented which are commonly detected in both BAGEL and antiSMASH tools, for

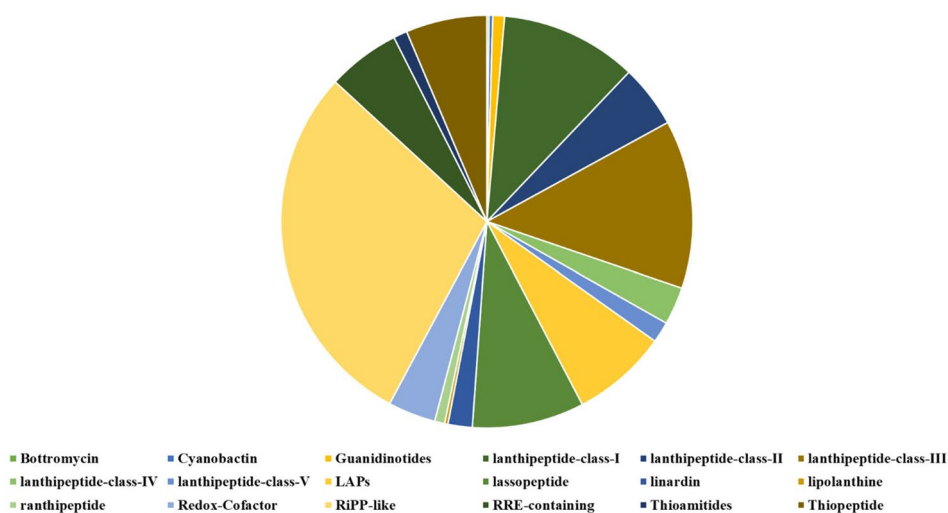


Figure 2. Distribution of different RiPPs (antiSMASH hits).

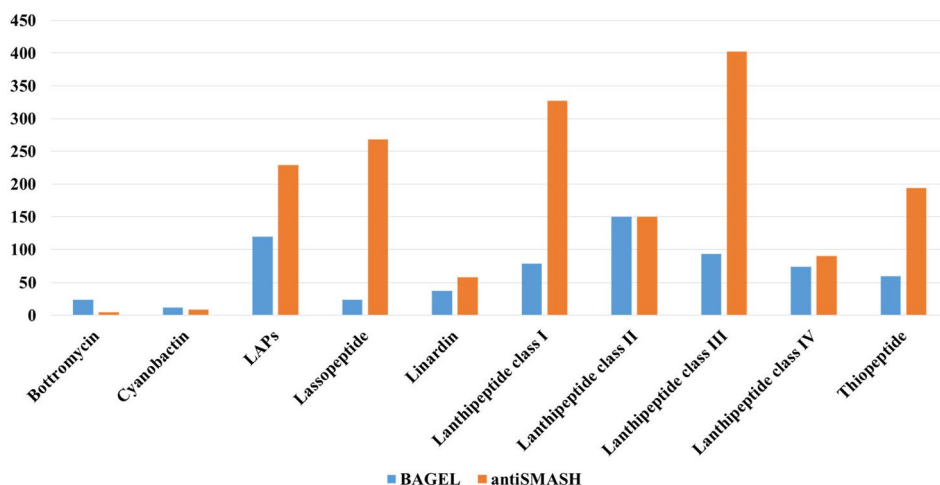


Figure 3. Comparison of RiPPs hits between BAGEL and antiSMASH genome mining tools.

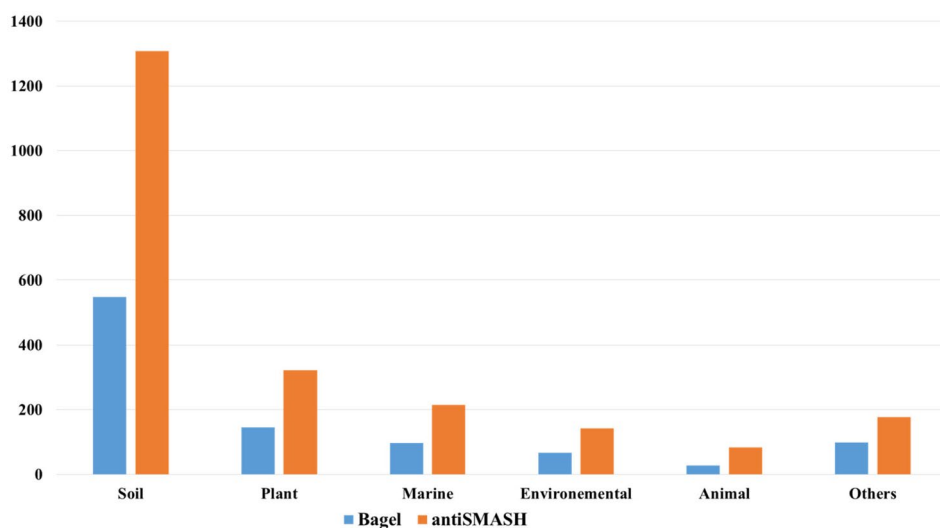


Figure 4. RiPPs hits based on sources according to BAGEL and antiSMASH.

instance, lanthipeptide-class I, lanthipeptide-class II, lanthipeptide-class III, lanthipeptide-class IV, Lap, lasso peptide, linaridin, and thiopeptide. In both analysis tools, every class of RiPPs is mainly identified in soil habitats. According to BAGEL analysis, lanthipeptide-class II is the most prevalent RiPPs type in soil habitat; on the other hand, it is lanthipeptide-class II according to antiSMASH analysis. Apart from soil habitat, the BAGEL tool detected a meager amount of RiPPs classes and other sources associated with *Streptomyces* bacteria. In comparison with the antiSMASH tool, significant numbers of RiPPs classes were identified in *Streptomyces* bacteria from other sources except animals associated with *Streptomyces* bacteria (Figure 5, Supplementary File-SF 6).

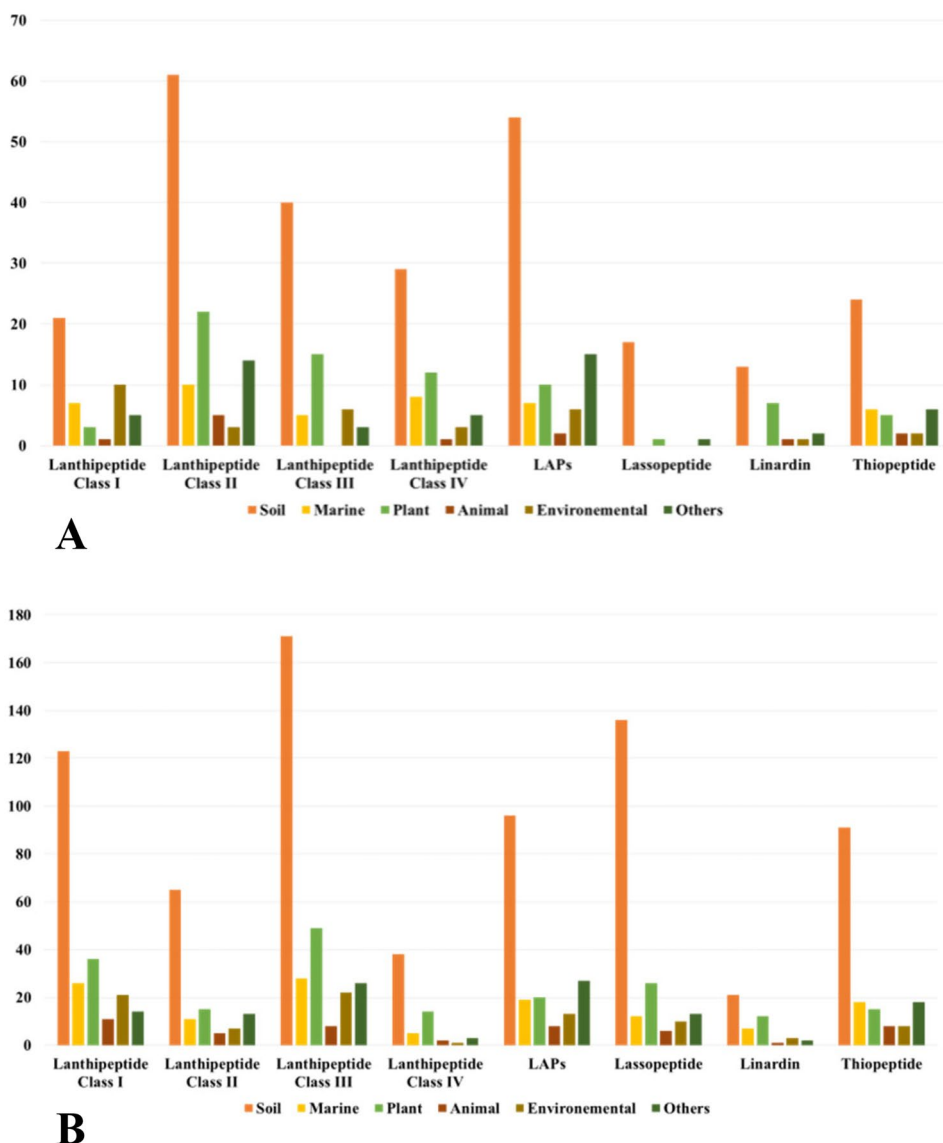


Figure 5. Habitat Sources data based on (A) BAGEL and (B) antiSMASH.

5. Discussion

Publicly available genome sequences of *Streptomyces* bacteria make genome mining research easy to pursue. As a result, a combination of different computational genome mining tools reveals a total of 4098 RiPP compounds discovered from only 410 *Streptomyces* strains. In contrast, bacteriocins and lanthipeptides are the most prevalent compounds. Most of the compounds produced in soil are associated *with* *Streptomyces* bacteria. Some of the RiPP compounds are not classified yet, but these secondary metabolites are promising targets for future research to explore their additional potential activities.

5.1. Bottromycins

In 1955, *Streptomyces bottropensis* bacteria from *Streptomyces* species were identified in soil samples and fermented in a broth, where researchers found out that showed promising anti-bacterial activity against multidrug resistance pathogens such as methicillin-resistant *Staphylococcus aureus* and vancomycin-resistant enterococci and led to the discovery of a new antibiotic, named bottromycins. Bottromycin is a macrocycle structured peptide which formed by a rare amidine linkage, C-methylation, and a D-amino acid [31].

The genome mining tool searched for the core peptide of bottromycins peptide, and both the tools performed to find out the novel bottromycins in the *Streptomyces* genome where BAGEL and antiSMASH detected 24 and 5 bottromycins, respectively. Soil and plant-associated *Streptomyces* show the most potential to produce bottromycins.

5.2. Cyanobactins

Cyanobacteria mostly produce cyanobactins, and researchers already identified around 200 cyanobactins. Even though traditional chemical methods mostly detect them, 1/3 were discovered by genome mining. However, cyanobactins are primarily found in cyanobacteria; some similar structures (telomestatin) are also denoted, which are produced from different species, such as *Streptomyces* [4].

This study uses BAGEL and antiSMASH to discover novel cyanobactins in the studied *Streptomyces* genome. However, BAGEL detected more cyanobactins; overall, a minimal amount was detected, mostly from soil habitats.

5.3. LAPs

LAPs are another type of RiPPs. Ser/Thr and Cys are converted to heterocyclic amino acids such as oxazoles and thiazoles via enzymatic cyclodehydration and dehydrogenation reactions. This kind of heterocyclic structural feature is unique in RiPPs classes, making LAPs distinguishable. The biosynthesis of LAPs is a bit of a complicated process because of the complex enzymes involved in this process. Streptolysin S (SLS) was the first LAP discovered from *Streptococcus pyogenes*, and SLS is a beta-hemolytic factor. But SLS was not an established LAP until the discovery of another LAP called microcin B17. After the classification of LAPs was established, a number of LAPs were discovered using the genome mining approach from publicly available genomes. All kinds of LAPs function against actinomycetes, including *Streptomyces scabies* JCM7914, a plant pathogen that causes the disease potato scab as potential peptide molecules [4].

A very good number of LAPs were identified in the studied *Streptomyces* genomes by both the genome mining tools, where antiSMASH discovered a higher number of LAPs than BAGEL with the addition of 109 hits, which resulted in almost 229 LAPs hits. A wide range of LAPs was identified in soil habitats, and a very low amount has been detected in animal-associated *Streptomyces* genome. On the other hand, a good number of LAPs hits were also observed in other categorized sources.

5.4. Lasso peptide

The name of lasso peptide is derived from its unique knotted structure called lasso fold, an emerging bacterial bioactive peptide that was enlightened in 1991 for the first time. A wide range of activities has been found in lasso peptides, such as antimicrobial, enzyme inhibitory, and receptor antagonistic activities. In 2008, selective characterization and isolation were applied by genome mining approach so that lasso peptide could be widely understood [32].

Lasso peptide detection in the studied *Streptomyces* genome gives a surprising amount of hits. Though BAGEL could have detected a minimal amount of lasso peptide, antiSMASH identified a total of 268 lasso peptide hits. All the lasso peptide are generated in all kinds of habitats but are mostly synthesized from soil-associated *Streptomyces* bacteria.

5.5. Linaridins

Linaridins were considered as crosslinks with the lanthipeptide family because of the presence of a C-terminal amino vinyl cysteine in Cypemycin structure, which is the first discovered peptide in the linaridin family. Later, the characterization of BGCs proved that lanthipeptides dehydratases of the four lanthipeptides classes are not present in the gene clusters. As a result, linaridin family is grouped separately and also considered a recently discovered family, which needs further research [4].

The genome mining approach applied to the studied *Streptomyces* genome sequence revealed the presence of linaridins family. However, there is a very small amount of linaridins detected; both the genome mining tools characterize almost similar amounts of linaridins, which are higher than bottromycins and cyanobactins. All linaridins are generated in every type of source associated with *Streptomyces* bacteria.

5.6. Lanthipeptide

Lanthipeptides are the largest group of the whole RiPPs family, and their genes are the most frequently found in bacteria. In lanthipeptides group, four types of unusual amino acid are produced due to their extensive PTM such as lanthionine (Lan), b-methylanthionine (MeLan), didehydroalanine (Dha; formed by dehydration of serine), and didehydrobutyrine (Dhb; formed by dehydration of threonine) [33, 34]. This unusual amino acid contributed in the unique and differentiated peptide structure of lanthipeptides where Dha or Dhb bind with a cysteine through thioether bridges and form Lan and MeLan. The strong conservation and modification of these enzymes, lanthipeptides, resulted in four different classes (I, II, III, and IV) [4]. Genome mining tools function by searching for specific ring-forming enzymes.

The lanthipeptides function as antimicrobial compounds, also known as lantibiotics. Lantibiotics are mostly produced by gram-positive bacteria and work against antibiotic-resistant and clinically relevant strains of *Staphylococcus*, *Enterococcus*, and *Clostridium* spp. Lantibiotics gain wide attention for clinical application because lantibiotics can bind to lipid-II, which cannot be inhibited by vancomycin. Since a complex enzymatic process produces lipid-II, it will be difficult to develop resistance

that characterizes lantibiotics as promising compounds to treat resistant bacterial disease [35–37].

Among the other RiPPs family, lanthipeptides are the most detected RiPPs family in the studied *Streptomyces* genome sequences. A total of 404 and 1019 hits were detected by BAGEL and antiSMASH genome mining tools, respectively. Among the massive lanthipeptides hits, only eight lanthipeptides hits are generated by BAGEL genome mining tool, which is not categorized in the lanthipeptides classes. An unusual hit has been generated by antiSMASH tool, which is lanthipeptide class V, although the number of lanthipeptide class V hits is only 50.

According to the BAGEL genome mining analysis, lanthipeptides class II is the most prevalent, and lanthipeptides class IV is the least one. All kind of lanthipeptides were detected in almost all kind of source-associated *Streptomyces* bacteria except the animal habitats, which contains a very small amount of lanthipeptides, especially lanthipeptides class III, which has not been detected in animal-associated *Streptomyces* bacteria.

On the other hand, lanthipeptide class III is the most prevalent one, which has generated 402 hits. A similar scenario was also observed in the case of lanthipeptides class I as in antiSMASH genome mining analysis. Only 50 hits of lanthipeptides class V have been generated, which is the least among the lanthipeptides' classified classes. Although soil is the most prevalent source of lanthipeptides biosynthesis, the rest of the habitats associated with bacteria also contribute to generating them, especially plant and marine source-based *Streptomyces* bacteria.

5.7. Thiopeptides

Thiopeptides were registered as a RiPPs family by the discovery of micrococcin in 1948 and have since converted into a family of around 100 metabolites. Thiopeptides are the most modified sulphur-rich peptides for their macrocycle containing thiazole rings with multiple dehydrated amino acid residues. As a result, thiopeptides are also known as thiazolyl peptides or pyridinyl polythiazolyl peptides. They share a series of common motifs that help to differentiate thiopeptides from other peptide-derived and azole-containing NPs. Most of the thiopeptides synthesis in Actinobacteria still Firmicutes, including those from the *Bacillus* and *Staphylococcus* genera, also featured for the biosynthesis of thiopeptides. Thiopeptides show several bioactivities, such as antimalarial and anticancer; however, they show more potential activity as anti-gram-positive bacterial peptides through interaction with bacterial protein synthesis. There are seven proteins corresponding to TpdA-G denoted in every cluster of thiopeptides to build the highly modified scaffold of a thiopeptide. The genome mining approach is also characterized as a suitable way to discover thiopeptides via targeting cyclodehydratase and dehydratase enzymes with the domains PF02624, PF00881, and PF14028 [4, 14].

A total of 194 thiopeptides are characterized by the genome mining tool antiSMASH. However, BAGEL can manage to identify only 60 hits of thiopeptides in the studied *Streptomyces* genome. This amount of thiopeptide hits shows a significant value in the biosynthesis of ribosomal peptides in comparison to other types of RiPPs. Although soil is the most prevalent source of thiopeptide biosynthesis, the

rest of the habitats associated with bacteria also contribute to generating them, especially marine source-based *Streptomyces* bacteria.

5.8. Other classes of RiPPs

RiPPs is a boarder family of peptides and contains a dozen classes. Both of the bioinformatics tools have detected some classes. There are nine other classes of RiPPs also detected such as bacteriocins, guanidinotides, lanthipeptide-class V, lipolanthine, ranthipeptide, redox-cofactor, Ripp-like, RRE-containing, thioamitides only by antiS-MASH tool except for bacteriocin which has been identified by BAGEL tool. All of the RiPPs classes have been identified in a very decent amount but the prevalence of RiPP-like and bacteriocin is very high. Almost 884 RiPP-like have been characterized, the highest number of RiPPs hits in this investigation. According to the BAGEL tool, bacteriocin also the highest amount of RiPPs classes among the others which is almost 354. On the other hand, RRE-containing and redox cofactors give a very significant amount of RiPPs hits such as 174 and 114 respectively as well. The rest of the RiPPs classes were detected very little, whereas lipolanthine was the lowest.

6. Conclusion

Many studies show that RiPPs have a high potential as bioactive NPs compared to other conventional NPs, which have failed to kill drug-resistant pathogens. Apart from the antimicrobial activity of RiPP compounds, they also have several other bioactivities against various diseases. As a result, RiPPs become a major research attraction in modern research society. However, traditional research methods can manage less than 10% of microorganism's biosynthetic capabilities discovered in search of bioactive NPs until the genome mining approach arrives. Public databases of genome sequences help researchers to detect bioactive NPs in rich bacterial species. *Streptomyces* species is a very well-known bacterial species for the production of heavy bioactive NPs [6]. However, there is some lack of evidence of the production of RiPPs by the *Streptomyces* bacterial strains due to a shortage of technology. Technological advancement opens the window to explore the RiPPs biosynthesis potential in the *Streptomyces* species. Notably, additional methodology and technology are required to classify the unknown RiPP compounds identified by the genome mining tools and further research on the validation of the activity of the individual RiPP compounds. Further study on the activation of down-regulating gene clusters can discover new types of potential RiPP compounds and open a new era in discovering secondary metabolites.

Streptomyces have sufficient NP substrates and sophisticated post-modification systems like phosphorylation, acetylation, farnesylation, and glycosylation that are necessary for the many actions of NPs; they make an ideal chassis for NP overexpression [38].

Disclosure statement

No interest declared.

Funding

This work was funded by the Bangladesh Council of Scientific & Industrial Research, Dhaka, Bangladesh (grant number: 24, 39.02.0000.011.14.128.2020/636, dated: 31.12.2020).

References

- [1] H.R. Karbalaeei-Heidari and N. Budisa, *Front. Microbiol.* **11**, 1 (2020).
- [2] R.E. de Lima Procópio, I.R. da Silva, M.K. Martins, J.L. de Azevedo, and J.M. de Araújo, *Braz. J. Infect. Dis.* **16**, 466 (2012).
- [3] C. Wu and W.A. van der Donk, *Curr. Opin. Biotechnol.* **69**, 221–231 (2021).
- [4] P.G. Arnison, M.J. Bibb, G. Bierbaum, A.A. Bowers, T.S. Bugni, G. Bulaj, J.A. Camarero, D.J. Campopiano, G.L. Challis, J. Clardy, P.D. Cotter, D.J. Craik, M. Dawson, E. Dittmann, S. Donadio, P.C. Dorrestein, K.D. Entian, M.A. Fischbach, J.S. Garavelli, U. Göransson, C.W. Gruber, D.H. Haft, T.K. Hemscheidt, C. Hertweck, C. Hill, A.R. Horswill, M. Jaspars, W.L. Kelly, J.P. Klinman, O.P. Kuipers, A.J. Link, W. Liu, M.A. Marahiel, D.A. Mitchell, G.N. Moll, B.S. Moore, R. Müller, S.K. Nair, I.F. Nes, G.E. Norris, B.M. Olivera, H. Onaka, M.L. Patchett, J. Piel, M.J. Reaney, S. Rebuffat, R.P. Ross, H.G. Sahl, E.W. Schmidt, M.E. Selsted, K. Severinov, B. Shen, K. Sivonen, L. Smith, T. Stein, R.D. Süßmuth, J.R. Tagg, G.L. Tang, A.W. Truman, J.C. Vederas, C.T. Walsh, J.D. Walton, S.C. Wenzel, J.M. Willey, and W.A. van der Donk, *Nat. Prod. Rep.* **30**, 108 (2013).
- [5] M.E. Hamid, A. Mahgoub, A.J.O. Babiker, H.A.E. Babiker, M.A.I. Holie, M.M. Elhassan, and M.R.P. Joseph, *Int. J. Environ. Res. Public Health.* **17**, 8749 (2020).
- [6] K. Alam, A. Mazumder, S. Sikdar, Y.M. Zhao, J. Hao, C. Song, Y. Wang, R. Sarkar, S. Islam, Y. Zhang, and A. Li, *Front. Microbiol.* **13**, 968053 (2022).
- [7] R. Kurnijasanti and S.A. Sudjarwo, *Vet. World* **15**, 634 (2022).
- [8] K. Alam, J. Hao, Y. Zhang, and A. Li, *Biotechnol. Adv.* **49**, 107759 (2021).
- [9] K. Alam, J. Hao, L. Zhong, G. Fan, Q. Ouyang, M.M. Islam, S. Islam, H. Sun, Y. Zhang, R. Li, and A. Li, *Front. Microbiol.* **13**, 939919 (2022).
- [10] K. Alam, M.M. Islam, J. Hao, M.N. Abbasi, M. Hayat, M. Shoaib, Y. Zhang, and A. Li, *World J. Microbiol. Biotechnol.* **39**, 64 (2023).
- [11] K. Alam, M.M. Islam, K. Gong, M.N. Abbasi, R. Li, Y. Zhang, and A. Li, *Comput. Biol. Med.* **140**, 105046 (2022).
- [12] Z. Zhong, B. He, J. Li, and Y.X. Li, *Synth. Syst. Biotechnol.* **5**, 155 (2020).
- [13] A.H. Russell and A.W. Truman, *Comput. Struct. Biotechnol. J.* **18**, 1838 (2020).
- [14] N. Poorinmohammad, R. Bagheban-Shemirani, and J. Hamed, *Anton. Leeuw. Int. J. Gen. Mol. Microbiol.* **112**, 1477 (2019).
- [15] P.D. Cotter, R.P. Ross, and C. Hill, *Nat. Rev. Microbiol.* **11**, 95 (2013).
- [16] M.H. Medema, K. Blin, P. Cimermancic, V. De Jager, P. Zakrzewski, M.A. Fischbach, T. Weber, E. Takano, and R. Breitling, *Nucleic Acids Res.* **39**, 339 (2011).
- [17] A.J. van Heel, A. de Jong, C. Song, J.H. Viel, J. Kok, and O.P. Kuipers, *Nucleic Acids Res.* **46**, W278 (2018).
- [18] S.F. Altschul, W. Gish, W. Miller, E.W. Myers, and D.J. Lipman, *J. Mol. Biol.* **215**, 403 (1990).
- [19] R.D. Finn, J. Mistry, J. Tate, P. Coggill, A. Heger, J.E. Pollington, O.L. Gavin, P. Gunasekaran, G. Ceric, K. Forslund, and L. Holm, *Nucleic Acids Res.* **38**, 211 (2010).
- [20] D.A. Benson, K. Clark, I. Karsch-Mizrachi, D.J. Lipman, J. Ostell, and E.W. Sayers, *Nucleic Acids Res.* **43**, D30 (2015).
- [21] A. Bateman, *Nucleic Acids Res.* **43**, D204 (2015).
- [22] R. Hammami, A. Zouhir, C.L. Lay, J.B. Hamida, and I. Fliss, *BMC Microbiol.* **10**, 1 (2010).
- [23] F.H. Wagh, R.S. Barai, P. Gurung, and S. Idicula-Thomas, *Nucleic Acids Res.* **44**, D1094 (2016).

- [24] M.H. Medema, R. Kottmann, P. Yilmaz, M. Cummings, J.B. Biggins, K. Blin, I. de Bruijn, Y.H. Chooi, J. Claesen, R.C. Coates, P. Cruz-Morales, S. Duddela, S. Dusterhus, D.J. Edwards, D.P. Fewer, N. Garg, C. Geiger, J.P. Gomez-Escribano, A. Greule, M. Hadjithomas, A.S. Haines, E.J.N. Helfrich, M.L. Hillwig, K. Ishida, A.C. Jones, C.S. Jones, K. Jungmann, C. Kegler, H.U. Kim, P. Kötter, D. Krug, J. Masschelein, A.V. Melnik, S.M. Mantovani, E.A. Monroe, M. Moore, N. Moss, H.-W. Nützmann, G. Pan, A. Pati, D. Petras, F.J. Reen, F. Rosconi, Z. Rui, Z. Tian, N.J. Tobias, Y. Tsunematsu, P. Wiemann, E. Wyckoff, X. Yan, G. Yim, F. Yu, Y. Xie, B. Aigle, A.K. Apel, C.J. Balibar, E.P. Balskus, F. Barona-Gómez, A. Bechthold, H.B. Bode, R. Borriss, S.F. Brady, A.A. Brakhage, P. Caffrey, Y.-Q. Cheng, J. Clardy, R.J. Cox, R. De Mot, S. Donadio, M.S. Donia, W.A. van der Donk, P.C. Dorrestein, S. Doyle, A.J.M. Driessen, M. Ehling-Schulz, K.-D. Entian, M.A. Fischbach, L. Gerwick, W.H. Gerwick, H. Gross, B. Gust, C. Hertweck, M. Höfte, S.E. Jensen, J. Ju, L. Katz, L. Kaysser, J.L. Klassen, N.P. Keller, J. Kormanec, O.P. Kuipers, T. Kuzuyama, N.C. Kyrpides, H.-J. Kwon, S. Lautru, R. Lavigne, C.Y. Lee, B. Linqun, X. Liu, W. Liu, A. Luzhetskyy, T. Mahmud, Y. Mast, C. Méndez, M. Metsä-Ketelä, J. Micklefield, D.A. Mitchell, B.S. Moore, L.M. Moreira, R. Müller, B.A. Neilan, M. Nett, J. Nielsen, F. O’Gara, H. Oikawa, A. Osbourn, M.S. Osburne, B. Ostash, S.M. Payne, J.-L. Pernodet, M. Petricek, J. Piel, O. Ploux, J.M. Raaijmakers, J.A. Salas, E.K. Schmitt, B. Scott, R.F. Seipke, B. Shen, D.H. Sherman, K. Sivonen, M.J. Smanski, M. Sosio, E. Stegmann, R.D. Süßmuth, K. Tahlan, C.M. Thomas, Y. Tang, A.W. Truman, M. Viaud, J.D. Walton, C.T. Walsh, T. Weber, G.P. van Wezel, B. Wilkinson, J.M. Willey, W. Wohlleben, G.D. Wright, N. Ziemert, C. Zhang, S.B. Zotchev, R. Breitling, E. Takano, and F.O. Glöckner, *Nat. Chem. Biol.* **11**, 625 (2015).
- [25] K. Blin, M.H. Medema, D. Kazempour, M.A. Fischbach, R. Breitling, E. Takano, and T. Weber, *Nucleic Acids Res.* **41**, W204 (2013).
- [26] T. Weber, K. Blin, S. Duddela, D. Krug, H.U. Kim, R. Bruccoleri, S.Y. Lee, M.A. Fischbach, R. Müller, W. Wohlleben, R. Breitling, E. Takano, and M.H. Medema, *Nucleic Acids Res.* **43**, W237 (2015).
- [27] K. Blin, T. Wolf, M.G. Chevrette, X. Lu, C.J. Schwalen, S.A. Kautsar, H.G. Suarez Duran, E.L.C. de Los Santos, H.U. Kim, M. Nave, J.S. Dickschat, D.A. Mitchell, E. Shelest, R. Breitling, E. Takano, S.Y. Lee, T. Weber, and M.H. Medema, *Nucleic Acids Res.* **45**, W36 (2017).
- [28] K. Blin, S. Shaw, K. Steinke, R. Villebro, N. Ziemert, S.Y. Lee, M.H. Medema, and T. Weber, *Nucleic Acids Res.* **47**, W81 (2019).
- [29] K. Blin, S. Shaw, A.M. Kloosterman, Z. Charlop-Powers, G.P. Van Wezel, M.H. Medema, and T. Weber, *Nucleic Acids Res.* **49**, W29 (2021).
- [30] S.A. Kautsar, H.G. Suarez Duran, K. Blin, A. Osbourn, and M.H. Medema, *Nucleic Acids Res.* **45**, W55 (2017).
- [31] L. Franz, U. Kasmaier, A.W. Truman, and J. Koehnke, *Nat. Prod. Rep.* **38**, 1659 (2021).
- [32] J.D. Hegemann, M. Zimmermann, X. Xie, and M.A. Marahiel, *Acc. Chem. Res.* **48**, 1909 (2015).
- [33] C. Li, K. Alam, Y. Zhao, J. Hao, Q. Yang, Y. Zhang, R. Li, and A. Li, *Front. Bioeng Biotechnol.* **9**, 692466 (2021).
- [34] G.A. Hudson, and D.A. Mitchell, *Curr. Opin. Microbiol.* **45**, 61 (2018).
- [35] K.M. Gomes, R.S. Duarte, M. do C. de, and F. Bastos, *Microbiology* **163**, 109–121. (2017)
- [36] L.I. Kononova, L.B. Filatova, D.V. Eroshenko, and V.P. Korobov, *Microbiology* **86**, 571 (2017).
- [37] L.A. Draper, P.D. Cotter, C. Hill, and R.P. Ross, *Microbiol. Mol. Biol.* **79**, 171 (2015).
- [38] R. Liu, Z. Deng, and T. Liu, *Metab. Eng.* **50**, 74 (2018).